

BRAC UNIVERSITY

Department of Mathematics and Physical Sciences

MAT110: Differential Calculus and Coordinate Geometry

Lecture 15

Polar Coordinates and Coordinate Transformations

Polar coordinates; relation between polar and rectangular coordinates; translation and rotation of axes; and the effect of translation on a general equation of second degree.

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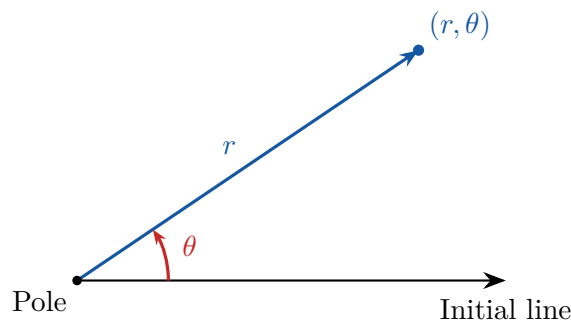
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1. Polar Coordinates

Polar Coordinates

In the polar coordinate system, a point is represented by an ordered pair (r, θ) , where r is the directed distance of the point from the **pole** and θ is the directed angle measured from the **initial line**.

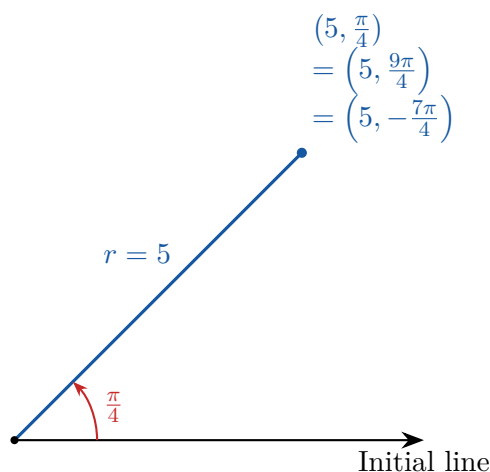


Example 01

Represent the polar coordinate $(r, \theta) = \left(5, \frac{\pi}{4}\right)$ and give coterminal representations of the same point.

Answer. The point lies five units from the pole on the ray $\theta = \pi/4$. Since adding or subtracting 2π does not change the terminal ray,

$$\left(5, \frac{\pi}{4}\right) = \left(5, 2\pi + \frac{\pi}{4}\right) = \left(5, \frac{9\pi}{4}\right) = \left(5, -\frac{7\pi}{4}\right).$$



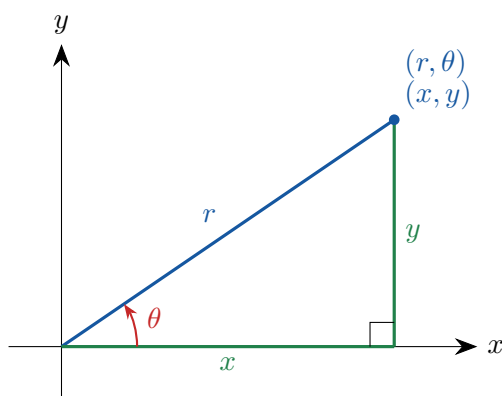
2. Relation Between Polar and Rectangular Coordinates

2.1 Polar to Rectangular Coordinates

Conversion from (r, θ) to (x, y)

For a point P with polar coordinates (r, θ) and rectangular coordinates (x, y) ,

$$\boxed{x = r \cos \theta}, \quad \boxed{y = r \sin \theta}.$$



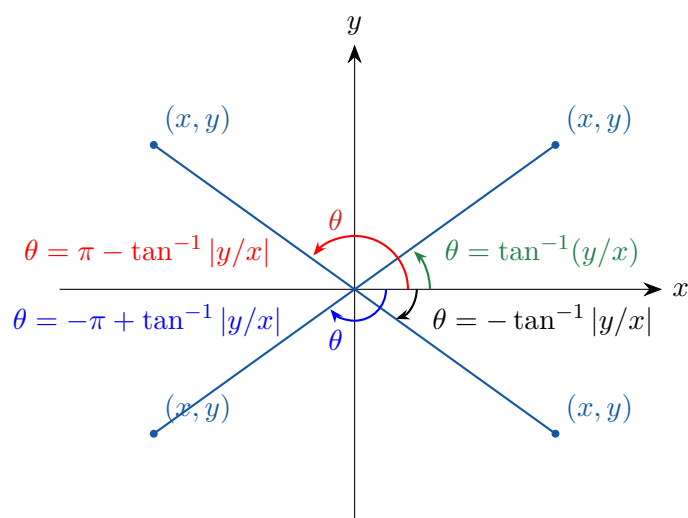
2.2 Rectangular to Polar Coordinates

Conversion from (x, y) to (r, θ)

The distance from the pole is

$$r = \sqrt{x^2 + y^2}.$$

For the standard range $-\pi < \theta \leq \pi$, the angle is selected according to the quadrant of (x, y) .



Example 02

Convert the point $(x, y) = (2\sqrt{3}, -2)$ to polar coordinates (r, θ) .

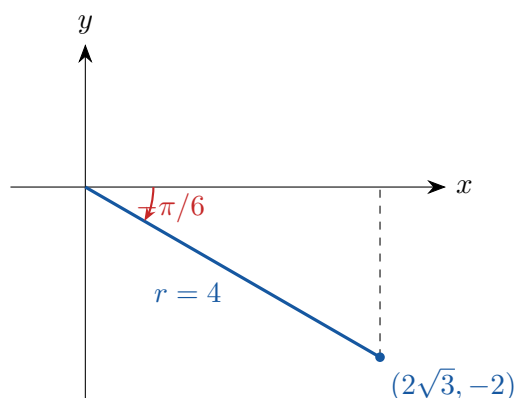
Answer. The point is in the fourth quadrant. Therefore,

$$r = \sqrt{(2\sqrt{3})^2 + (-2)^2} = \sqrt{12 + 4} = 4,$$

$$\theta = -\tan^{-1} \left| \frac{-2}{2\sqrt{3}} \right| = -\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = -\frac{\pi}{6}.$$

Hence,

$$(r, \theta) = \left(4, -\frac{\pi}{6} \right).$$

**Example 03**

Write $9x^2 + 8y^2 = 25$ in polar coordinates.

Answer. Using $x = r \cos \theta$ and $y = r \sin \theta$,

$$\begin{aligned} 9x^2 + 8y^2 &= 25, \\ 9r^2 \cos^2 \theta + 8r^2 \sin^2 \theta &= 25, \\ 8r^2(\cos^2 \theta + \sin^2 \theta) + r^2 \cos^2 \theta &= 25, \\ 8r^2 + r^2 \cos^2 \theta &= 25, \\ r^2(8 + \cos^2 \theta) &= 25. \end{aligned}$$

Thus,

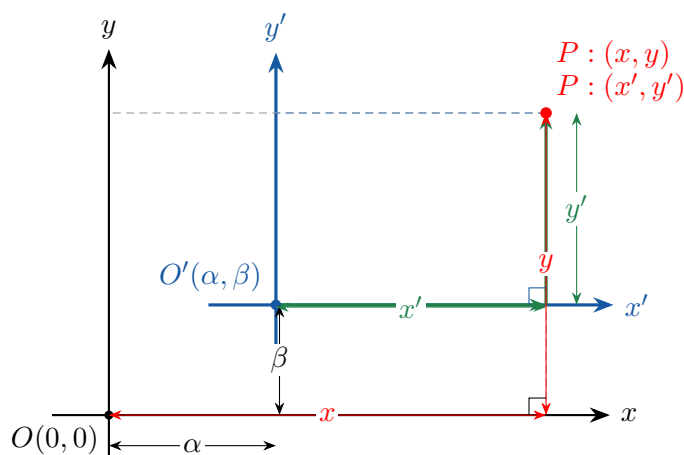
$$r^2 = \frac{25}{8 + \cos^2 \theta}.$$

3. Transformation of Coordinates**3.1 Translation Transformation****Translation of Axes**

Suppose the origin is translated from $O(0, 0)$ to $O'(\alpha, \beta)$ while the new axes remain parallel to the old axes. If a point has coordinates (x, y) in the old system and (x', y') in the new system, then

$$\boxed{x = \alpha + x'}, \quad \boxed{y = \beta + y'}.$$

Equivalently, $x' = x - \alpha$ and $y' = y - \beta$.



$$\begin{aligned} x &= \alpha + x' \\ y &= \beta + y' \end{aligned}$$

Example 04

Transform

$$(x - 2)^2 + (y - 3)^2 = 5^2$$

to a new frame of reference whose origin is at $(2, 3)$ and whose new axes are parallel to the old axes.**Answer.** Here $\alpha = 2$ and $\beta = 3$, so

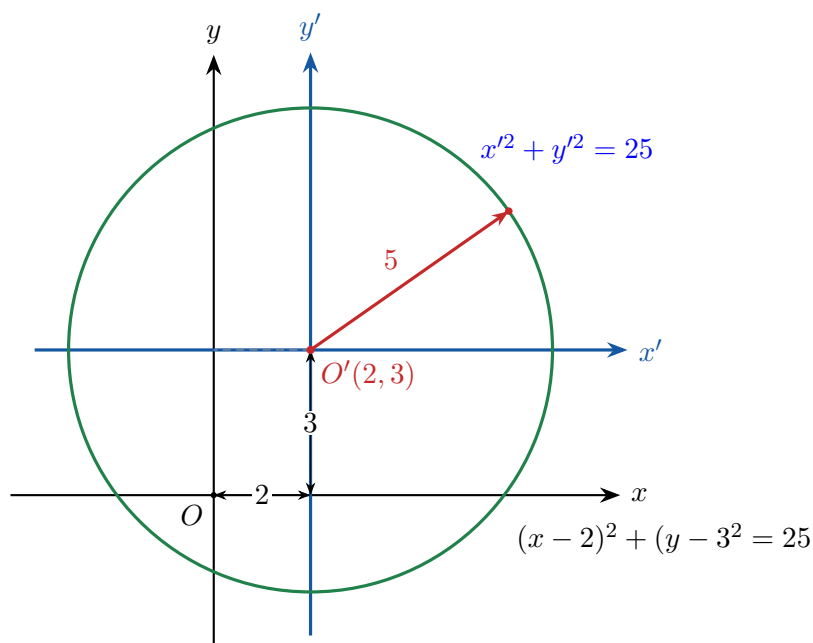
$$x = 2 + x', \quad y = 3 + y'.$$

Substituting into the given equation,

$$\begin{aligned} (2 + x' - 2)^2 + (3 + y' - 3)^2 &= 5^2, \\ x'^2 + y'^2 &= 5^2. \end{aligned}$$

Therefore, in the new frame of reference,

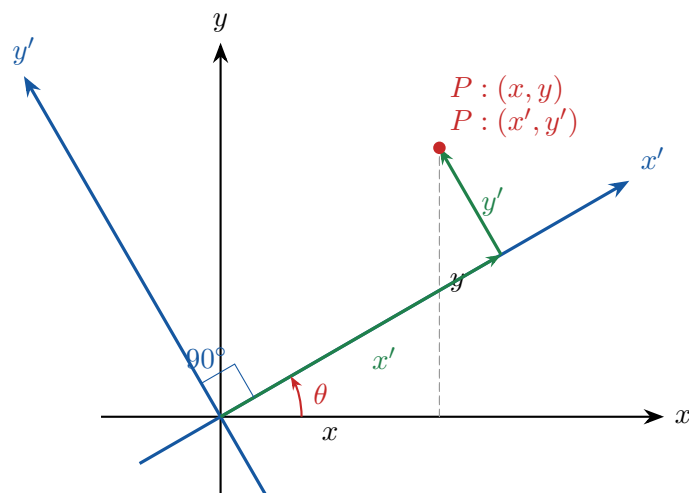
$$\boxed{x'^2 + y'^2 = 25}.$$

**3.2 Rotational Transformation****Rotation of Axes**Suppose the x' -axis is obtained by rotating the positive x -axis through an angle θ while the origin remains fixed. Then

$$\boxed{x = x' \cos \theta - y' \sin \theta}, \quad \boxed{y = x' \sin \theta + y' \cos \theta}.$$

In matrix form,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}.$$

**Example 05**

Transform the equation

$$x^2 - y^2 = a^2$$

to axes inclined at $\pi/4$ to the original axes.

Answer. For $\theta = \pi/4$,

$$x = x' \cos \frac{\pi}{4} - y' \sin \frac{\pi}{4} = \frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}},$$

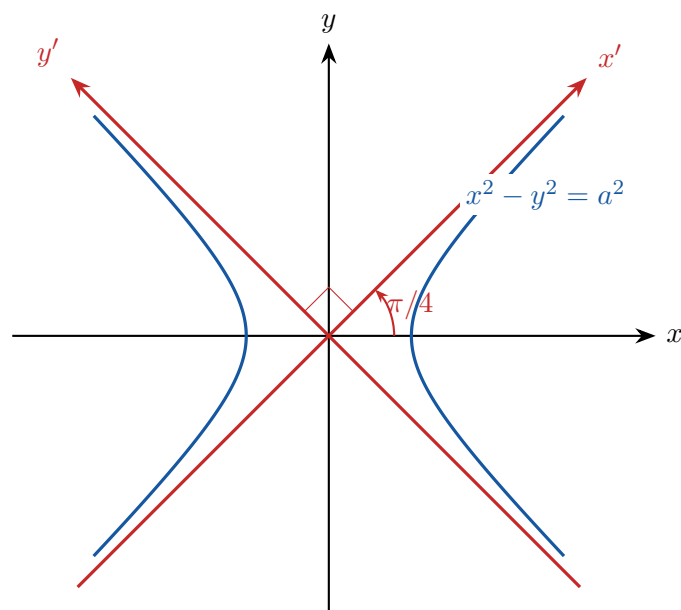
$$y = x' \sin \frac{\pi}{4} + y' \cos \frac{\pi}{4} = \frac{x'}{\sqrt{2}} + \frac{y'}{\sqrt{2}}.$$

Therefore,

$$\begin{aligned} x^2 - y^2 &= a^2, \\ \left(\frac{x'}{\sqrt{2}} - \frac{y'}{\sqrt{2}} \right)^2 - \left(\frac{x'}{\sqrt{2}} + \frac{y'}{\sqrt{2}} \right)^2 &= a^2, \\ 2x'y' &= a^2. \end{aligned}$$

Hence the transformed equation is

$$y' = \frac{a^2}{2x'}.$$



4. Effect of Translation on the General Equation of Second Degree

Consider the general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0. \quad (1)$$

Let (α, β) be the new origin and let the new x' - and y' -axes be parallel to the old x - and y -axes. Then

$$x = \alpha + x', \quad y = \beta + y'.$$

Substituting these expressions into (1),

$$\begin{aligned} a(\alpha + x')^2 + 2h(\alpha + x')(\beta + y') + b(\beta + y')^2 \\ + 2g(\alpha + x') + 2f(\beta + y') + c = 0. \end{aligned} \quad (2)$$

After expansion and collection of like terms,

$$\begin{aligned} ax'^2 + 2hx'y' + by'^2 + 2(a\alpha + h\beta + g)x' + 2(h\alpha + b\beta + f)y' \\ + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0. \end{aligned}$$

Write the transformed equation as

$$a'x'^2 + 2h'x'y' + b'y'^2 + 2g'x' + 2f'y' + c' = 0. \quad (3)$$

Comparing coefficients gives

$$\begin{aligned} a' &= a, & h' &= h, & b' &= b, \\ g' &= a\alpha + h\beta + g, & f' &= h\alpha + b\beta + f, \\ c' &= a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c. \end{aligned}$$

Let

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c.$$

Then

$$\begin{aligned} F(\alpha, \beta) &= a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = c', \\ \frac{\partial F}{\partial \alpha} &= 2a\alpha + 2h\beta + 2g = 2(a\alpha + h\beta + g) = 2g', \\ \frac{\partial F}{\partial \beta} &= 2h\alpha + 2b\beta + 2f = 2(h\alpha + b\beta + f) = 2f'. \end{aligned}$$

Thus,

$$g' = \frac{1}{2} \frac{\partial F}{\partial \alpha}, \quad f' = \frac{1}{2} \frac{\partial F}{\partial \beta}.$$

Transformed Equation under Translation

After translating the origin to (α, β) , the transformed equation is

$$ax'^2 + 2hx'y' + by'^2 + \frac{\partial F}{\partial \alpha}x' + \frac{\partial F}{\partial \beta}y' + F(\alpha, \beta) = 0.$$

The following quantities are invariant under translation:

- (i) $a' + b' = a + b$,
- (ii) $a'b' - h'^2 = ab - h^2$,
- (iii) $\Delta' = \Delta$, where

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}.$$

From the transformed equation, the x' - and y' -terms will be absent if

$$\boxed{\frac{\partial F}{\partial \alpha} = 0, \quad \frac{\partial F}{\partial \beta} = 0.}$$

Example 06

Transform the equation

$$9x^2 + 15xy + y^2 + 12x - 11y - 5 = 0$$

so as to remove the x - and y -terms.

Answer. Let

$$F(x, y) = 9x^2 + 15xy + y^2 + 12x - 11y - 5.$$

Let (α, β) be the new origin. To remove the linear terms, require

$$\begin{aligned} \frac{\partial F}{\partial \alpha} &= 18\alpha + 15\beta + 12 = 0, \\ \frac{\partial F}{\partial \beta} &= 15\alpha + 2\beta - 11 = 0. \end{aligned}$$

Solving these simultaneous equations gives

$$\alpha = 1, \quad \beta = -2.$$

Also,

$$\begin{aligned} c' = F(1, -2) &= 9(1)^2 + 15(1)(-2) + (-2)^2 + 12(1) - 11(-2) - 5 \\ &= 9 - 30 + 4 + 12 + 22 - 5 = 12. \end{aligned}$$

After translating the origin to $(1, -2)$, the transformed equation is

$$\boxed{9x'^2 + 15x'y' + y'^2 + 12 = 0}.$$

5. Problem Set

1. Find the polar co-ordinates of the points $(2\sqrt{3}, -2)$, $(0, -2)$, $(1, 1)$.
2. Find the rectangular co-ordinates of the points $(7, \frac{2\pi}{3})$, $(8, \frac{9\pi}{4})$, $(0, \pi)$.
3. Change to Cartesian coordinates the equations (i) $r = a \sin \theta$, (ii) $\sqrt{r} = \sqrt{a} \cos \left(\frac{\theta}{2}\right)$.
4. Transform to polar coordinates the equations (i) $9x^2 + 4y^2 = 36$, (ii) $x^3 = y^2(2a - x)$.
5. Transform to parallel axes through the new origin $(1, -2)$ the equation $2x^2 + y^2 - 4x + 4y = 0$.
6. Transform the equation $x^2 + y^2 - 8x + 14y + 5 = 0$ to parallel axes through $(4, -7)$.
7. Transform the equation $7x^2 - 2xy + y^2 + 1 = 0$ to axes turned through the angle $\tan^{-1}\left(\frac{1}{2}\right)$.
8. Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}\left(\frac{4}{3}\right)$.
9. Transform the equation $9x^2 + 15xy + y^2 + 12x - 11y - 5 = 0$ so as to remove the terms in x and y .
10. Transform the equation $11x^2 + 3xy + 7y^2 + 19 = 0$ so as to remove the term xy .
11. Determine the equation of the curve $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$ when the origin is transferred to the point $(-2, 3)$.
12. Remove the xy -term from the equation $9x^2 + 24xy + 2y^2 + 54 = 0$.
13. Determine the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$ after rotating the axes through 30° .
14. Transform the equation $9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0$ so as to remove the terms in x and y .